Clovis SFEIR

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Profile

IMT Atlantique engineering student specialising in Python data pipelines and production backend systems. Shipped a production RAG pipeline (Docker, GCP) cutting lookup time **40%**, and built five personal systems (Python/C++20/Rust) benchmarked against industry references or labeled ground-truth, including a from-scratch retrieval engine (**MRR 0.944**) and an async telemetry pipeline processing **1.6M samples/s** with **0/10 false positives**.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*

Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France ***Class rank: 1/85** | Software Engineering, Databases, Cybersecurity*

HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlAce (1.458); field includes Stanford, Princeton, ETH Zürich.

Work Experience

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- Production RAG data pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking), deployed with **Docker** on **GCP**: **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana monitoring.
- Real-time interview simulator (Next.js/React, TypeScript, GPT-4o, ElevenLabs TTS): streaming responses over a scenario-branching state machine, backed by a Python API layer.

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis, epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair; 928 Rust tests.

Projects

Engineering Docs RAG *Python, BM25, information retrieval*

- From-scratch Okapi BM25 retrieval engine: chunking, ranking, and a pluggable extractive/LLM generator interface, built as a data pipeline from raw documents to ranked, served results – zero paid API calls in tests/CI/demo by design.
- Hand-derived BM25 scoring verified to 9 significant figures (TDD-first, not tuned by trial and error); measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; ~3,330 docs/s ingest; 53 tests, zero runtime dependencies.

IIoT Predictive Maintenance *Rust, tokio, EWMA/CUSUM*

- Async telemetry data pipeline (`tokio` mpsc, backpressure-aware channels) fuses EWMA + CUSUM control charts across 3 concurrent sensor streams to detect developing degradation signatures in real time.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818; **1.6 M samples/s**; 26 tests, 0 failures.

Crude Oil Spread Trading Engine *C++20, stochastic processes, order book*

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9}; 58 assertions/21 tests, ASan+UBSan clean.

HFT Limit Order Book Matching Engine *C++20, cache-aware layout, intrusive data structures*

- Price-time-priority matching engine (limit/market/cancel) with intrusive doubly-linked price-level lists (**O(1) cancel** via pointer unlink) and pre-allocated object pooling.
- Benchmark suite across three synthetic order-flow regimes: **2.3M+ msg/s** single-threaded throughput, **P50 0.6–1.0 µs** latency, zero hot-path allocations; unit tested.

Technical Skills

Languages	Python, SQL, TypeScript, C++20, Rust
Data / Infra	Docker, GCP, tokio async, Prometheus/Grafana, GitHub Actions CI, Redis, ZeroMQ
ML / Retrieval	BM25/information retrieval, EWMA/CUSUM change-detection, Kalman filter, Ornstein-Uhlenbeck
Perf / Systems	cache-aware layout, intrusive linked lists, acquire/release atomics, ICS/SCADA protocol parsing
Frameworks	Flask, Node.js/Express, React/Next.js, REST/gRPC